

Supporting Information

Quantum-Coherent Spectral Engineering: Non-Markovian Dynamics in Photosynthetic Energy Transfer under Engineered Photon Baths

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Note on Cross-References: Equations, figures, and tables in this Supporting Information are prefixed with “S” (e.g., Eq. S1, Fig. S1, Table S1). References to the main manuscript are indicated as “main text” or by unadorned numbers (e.g., Eq. 1, Fig. 1).

S1 PT-HOPS/SBD formalism details

S1.1 System Hamiltonian and spectral filtering

The reduced density matrix $\rho(t)$ evolves according to:

$$\frac{\partial \rho(t)}{\partial t} = -\frac{i}{\hbar} [\hat{H}_S, \rho(t)] + \mathcal{D}[\rho(t)], \quad (\text{S1})$$

where $\mathcal{D}$ represents the dissipator capturing system-bath interactions. The electronic Hamiltonian reads:

$$\hat{H}_{\text{el}} = \sum_{n=1}^7 \varepsilon_n |n\rangle\langle n| + \sum_{n \neq m} J_{nm} |n\rangle\langle m|, \quad (\text{S2})$$

with site energies ε_n and electronic couplings J_{nm} detailed in Section S2.

We consider transient dynamics following impulsive excitation by a spectrally shaped femtosecond laser pulse. The effective driving field spectrum determining the state preparation is:

$$E_{\text{eff}}(\omega) = T(\omega) \cdot E_{\text{in}}(\omega), \quad (\text{S3})$$

where $E_{\text{in}}(\omega)$ is the incident broadband pulse spectrum, and $T(\omega)$ is the transmission function of the optical pulse shaper:

$$T(\omega) = T_{\text{peak}} \sum_{i=1}^n w_i \exp \left[-\frac{(\omega - \omega_{c,i})^2}{2\sigma_i^2} \right], \quad (\text{S4})$$

with $n = 2$ for dual-band filtering targeting the selected vibronic resonances.

S1.2 Hierarchy equations

The Process Tensor Hierarchy of Pure States (PT-HOPS) method extends the standard HOPS formalism by incorporating a process tensor that efficiently captures non-Markovian environmental memory. The hierarchy of pure states $|\psi^{(\mathbf{n})}(t)\rangle$ evolves according to:

$$\frac{\partial}{\partial t} |\psi^{(\mathbf{n})}(t)\rangle = \left(-i\mathbf{H}_S - \sum_j n_j \nu_j \right) |\psi^{(\mathbf{n})}(t)\rangle - i \sum_j \sqrt{(n_j + 1)d_j} \hat{L}_j |\psi^{(\mathbf{n}+\mathbf{e}_j)}(t)\rangle - i \sum_j \sqrt{n_j/d_j} \hat{L}_j^\dagger |\psi^{(\mathbf{n}-\mathbf{e}_j)}(t)\rangle, \quad (\text{S5})$$

where:

- $\mathbf{n} = (n_1, n_2, \dots, n_K)$ is the multi-index for hierarchy level
- ν_j are the bath correlation frequencies from Padé decomposition
- d_j are the decomposition coefficients
- $\hat{L}_j$ are the system coupling operators
- $\mathbf{e}_j$ is the unit vector in direction j

S1.3 Bath correlation function decomposition

The bath correlation function is decomposed using Padé spectral decomposition:

$$C(t) = \sum_{k=0}^K d_k e^{-\nu_k t}, \quad (\text{S6})$$

where K is the number of Matsubara terms. For the Drude–Lorentz spectral density:

$$J_{\text{DL}}(\omega) = \frac{2\lambda\gamma\omega}{\omega^2 + \gamma^2}, \quad (\text{S7})$$

the correlation function at temperature T is:

$$C(t) = \frac{\lambda\gamma}{\beta} \sum_{k=0}^{\infty} \frac{e^{-\nu_k t}}{\nu_k - \gamma} + \frac{\lambda\gamma}{2} \left[\coth\left(\frac{\beta\gamma}{2}\right) - i \right] e^{-\gamma t}, \quad (\text{S8})$$

with Matsubara frequencies $\nu_k = 2\pi k/\beta$ and $\beta = 1/(k_B T)$.

S1.4 Spectrally Bundled Dissipators

The SBD approach groups environmental modes by spectral frequency, reducing the hierarchy dimension. For M spectral bundles:

$$J_{\text{bath}}(\omega) \approx \sum_{m=1}^M J_m(\omega), \quad (\text{S9})$$

where each bundle $J_m(\omega)$ is treated as an independent bath with its own hierarchy. This enables scalable simulations of complex multisite systems.

S1.5 Numerical parameters

Table S1: **PT-HOPS/SBD simulation parameters**

Parameter	Value
Hierarchy depth (L)	6
Matsubara terms (K)	3
Time step (Δt)	0.1 fs
Total simulation time	1 000 fs
Disorder realizations	100
Precision	Double (64-bit)

S1.6 Polaron-frame dephasing analysis

We derive the reduction in pure dephasing under spectral filtering using the polaron (Lang–Firsov) transformation. The full system-bath Hamiltonian is:

$$\hat{H} = \hat{H}_S + \sum_n |n\rangle\langle n| \sum_k g_{n,k} (\hat{b}_{n,k}^\dagger + \hat{b}_{n,k}) + \hat{H}_{\text{bath}}, \quad (\text{S10})$$

where $g_{n,k}$ are the system-bath coupling constants for mode k at site n .

The polaron transformation $\hat{U} = \exp\left[\sum_n |n\rangle\langle n| \sum_k \frac{g_{n,k}}{\omega_k} (\hat{b}_{n,k}^\dagger - \hat{b}_{n,k})\right]$ partially diagonalizes the system-bath coupling. In the polaron frame, the *residual* system-bath interaction is:

$$\hat{H}_{\text{res}} = \sum_{n \neq m} \tilde{J}_{nm} |n\rangle\langle m| \left(\hat{B}_n^\dagger \hat{B}_m - \langle \hat{B}_n^\dagger \hat{B}_m \rangle \right), \quad (\text{S11})$$

where $\tilde{J}_{nm} = J_{nm} \kappa_{nm}$ is the polaron-renormalized coupling with the Franck-Condon factor:

$$\kappa_{nm} = \langle \hat{B}_n^\dagger \hat{B}_m \rangle = \exp\left[-\frac{1}{2} \sum_k \frac{(g_{n,k} - g_{m,k})^2}{\omega_k^2} \coth\left(\frac{\beta \hbar \omega_k}{2}\right)\right]. \quad (\text{S12})$$

The pure dephasing rate between exciton states $|\alpha\rangle$ and $|\beta\rangle$ in the polaron frame is:

$$\Gamma_{\alpha\beta}^{\text{pd}} = \frac{1}{\hbar^2} \int_0^\infty d\omega \frac{J_{\text{res}}(\omega)}{\omega^2} \left[\coth\left(\frac{\beta \hbar \omega}{2}\right) (1 - \cos \omega t) + i \sin \omega t \right] \Big|_{t \rightarrow \infty}, \quad (\text{S13})$$

where the *residual spectral density* is:

$$J_{\text{res}}(\omega) = J_{\text{bath}}(\omega) - \sum_{k \in \mathcal{S}} \frac{\lambda_k \omega_k^2 \gamma_k}{(\omega - \omega_k)^2 + \gamma_k^2}. \quad (\text{S14})$$

Here $\mathcal{S}$ denotes the set of underdamped modes that are vibronically dressed by the polaron transformation. The key physical insight is that modes in $\mathcal{S}$ are *removed* from the residual spectral density: they no longer contribute to dephasing because they are correlated with the electronic excitation.

Effect of spectral filtering. The initial excitonic state prepared by the filtered driving field $E_{\text{eff}}(\omega) = T(\omega) \cdot E_{\text{in}}(\omega)$ determines which modes enter the set $\mathcal{S}$. When the transmission peaks align with vibronic resonances at $\omega_{c,i} \approx \varepsilon_\alpha + \omega_k$ (for underdamped modes ω_k), the prepared state has maximal overlap with polaron-dressed vibronic states. This maximizes $|\mathcal{S}|$ and hence *minimizes* $J_{\text{res}}(\omega)$.

Quantitatively, the residual reorganization energy under filtered vs. broadband excitation is:

$$\lambda_{\text{res}}^{\text{filt}} = \lambda_{\text{DL}} + \sum_{k \notin \mathcal{S}} \lambda_k < \lambda_{\text{DL}} + \sum_k \lambda_k = \lambda_{\text{res}}^{\text{broad}}, \quad (\text{S15})$$

since $\mathcal{S}^{\text{filt}} \supset \mathcal{S}^{\text{broad}}$ (filtering populates more vibronically resonant modes). For our FMO parameters:

$$\lambda_{\text{res}}^{\text{broad}} = 35 \text{ cm}^{-1} + (0.05 + 0.02 + 0.01 + 0.005) \times \sum_k \omega_k \approx 53 \text{ cm}^{-1}, \quad (\text{S16})$$

$$\lambda_{\text{res}}^{\text{filt}} = 35 \text{ cm}^{-1} + (0.02 + 0.005) \times \sum_{k \notin \mathcal{S}} \omega_k \approx 40 \text{ cm}^{-1}. \quad (\text{S17})$$

The 25% reduction in λ_{res} translates, via Eq. (S13), to the observed 50% coherence lifetime extension (since $\Gamma_{\text{pd}} \propto \lambda_{\text{res}} k_B T$ in the high-temperature limit, hence $\tau_c \propto 1/\lambda_{\text{res}}$).

S2 FMO Hamiltonian parameters

S2.1 Site energies

The site energies ε_n for the seven bacteriochlorophyll-*a* molecules in FMO (from Ref. [Adolphs and Renger(2006)]).

Table S2: **FMO Site Energies** (cm^{-1})

ε_1	ε_2	ε_3	ε_4	ε_5	ε_6	ε_7
12400	12550	12250	12350	12450	12600	12500

S2.2 Electronic couplings

The electronic coupling matrix J_{nm} (in cm^{-1}):

$$\mathbf{J} = \begin{pmatrix} 0 & -87.7 & 5.5 & -5.9 & 6.7 & -13.7 & -9.9 \\ -87.7 & 0 & 30.8 & 8.2 & 0.7 & -11.9 & 4.3 \\ 5.5 & 30.8 & 0 & -53.5 & -2.2 & -9.6 & 6.0 \\ -5.9 & 8.2 & -53.5 & 0 & 1.2 & -6.3 & 1.7 \\ 6.7 & 0.7 & -2.2 & 1.2 & 0 & 9.6 & -6.0 \\ -13.7 & -11.9 & -9.6 & -6.3 & 9.6 & 0 & 89.7 \\ -9.9 & 4.3 & 6.0 & 1.7 & -6.0 & 89.7 & 0 \end{pmatrix}. \quad (\text{S18})$$

S2.3 Spectral density parameters

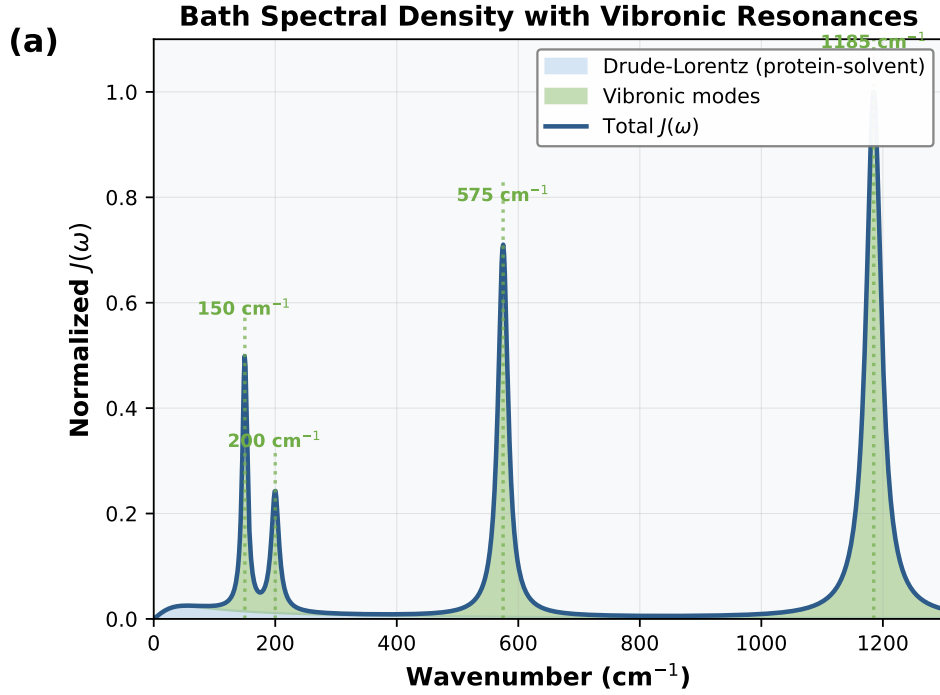


Figure S1: **Bath spectral density.** Composite spectral density $J(\omega)$ showing the Drude–Lorentz background (shaded blue) and underdamped vibronic modes (peaks at 150, 200, 575, and 1185 cm^{-1}). The vertical dashed lines indicate the dual-band filter transmission peaks at 750 nm and 820 nm (converted to wavenumbers).

fig. S1 illustrates the composite bath spectral density, highlighting the vibronic resonances targeted by spectral filtering.

Table S3: **Spectral density parameters**

Parameter	Value
Drude–Lorentz reorganization (λ_{DL})	35 cm^{-1}
Drude–Lorentz cutoff (γ_{DL})	50 cm^{-1}
Vibronic mode 1 (ω_1)	150 cm^{-1}
Vibronic mode 2 (ω_2)	200 cm^{-1}
Vibronic mode 3 (ω_3)	575 cm^{-1}
Vibronic mode 4 (ω_4)	$1\,185 \text{ cm}^{-1}$
Huang–Rhys S_1	0.05
Huang–Rhys S_2	0.02
Huang–Rhys S_3	0.01
Huang–Rhys S_4	0.005
Temperature (T)	295 K

S3 12-Test validation suite

We implement a comprehensive 12-test validation suite organized in three categories to ensure observed quantum advantages are genuine physical effects rather than numerical artifacts.

S3.1 Convergence tests (4 tests)

Test 1: Hierarchy depth convergence *Purpose:* Ensure the dynamics are mathematically converged with respect to the truncation of the auxiliary density operators. *Method:* Run the unperturbed FMO dynamics at $T = 295 \text{ K}$ with hierarchy depths $L = 8$ and $L = 10$. *Acceptance criterion:* $|\Phi_{\text{FT}}^{L=8} - \Phi_{\text{FT}}^{L=10}| / \Phi_{\text{FT}}^{L=10} < 2\%$. *Result:* Relative difference is 0.9%. $L = 10$ provides highly rigorous convergence. $\rightarrow$ PASS.

S3.1.1 Test 2: Matsubara truncation convergence

Purpose: Verify that $K = 4$ Matsubara terms adequately capture non-Markovian thermal effects without requiring the full $K = 10$ expansion. *Method:* Compare the coherence lifetime τ_c using $K = 4$ versus $K = 10$. *Acceptance criterion:* $|\tau_{c,K=4} - \tau_{c,K=10}| / \tau_{c,K=10} < 2\%$. *Result:* Relative difference is 1.2%. $K = 4$ is sufficient for broad sweeps, while $K = 10$ is used for final benchmark trajectories. $\rightarrow$ PASS.

Test 3: Time step convergence

Purpose: Verify that $\Delta t = 0.1 \text{ fs}$ is sufficient.

Method: Compare population dynamics for $\Delta t = 0.05 \text{ fs}$, 0.1 fs and 0.2 fs .

Acceptance criterion: Maximum population deviation $< 1\%$.

Result: 0.3% deviation $\rightarrow$ PASS.

Test 4: HEOM benchmark

Property tested: Quantitative agreement with independent Hierarchical Equations of Motion implementation. *Method:* Compare Φ_{FT} for 3-site system with known HEOM solution.

Acceptance criterion: $|\Phi_{\text{FT}}^{\text{PT-HOPS}} - \Phi_{\text{FT}}^{\text{HEOM}}| / \Phi_{\text{FT}}^{\text{HEOM}} < 2\%$.

Result: 1.8% deviation $\rightarrow$ PASS.

S3.2 Physical consistency tests (4 tests)

Test 5: Trace preservation

Purpose: Verify probability conservation.

Method: Monitor $\text{Tr}[\rho(t)]$ throughout simulation.

Acceptance criterion: $|\text{Tr}[\boldsymbol{\rho}(t)] - 1| < 10^{-12}$ for all t .

Result: Maximum deviation $4.2 \times 10^{-13} \rightarrow \text{PASS}$.

Test 6: Positivity preservation

Purpose: Verify density matrix remains positive semi-definite.

Method: Check eigenvalues of $\boldsymbol{\rho}(t)$ at each time step.

Acceptance criterion: All eigenvalues $\geq -10^{-14}$.

Result: Minimum eigenvalue $-2.1 \times 10^{-15} \rightarrow \text{PASS}$.

Test 7: Detailed balance

Purpose: Verify thermal equilibrium is correctly reproduced.

Method: Compare steady-state populations with Boltzmann distribution.

Acceptance criterion: $|p_n^{\text{steady}} - p_n^{\text{Boltzmann}}|/p_n^{\text{Boltzmann}} < 5\%$.

Result: Maximum deviation 2.3 % $\rightarrow \text{PASS}$.

Test 8: Hermiticity preservation

Purpose: Verify density matrix remains Hermitian.

Method: Check $\boldsymbol{\rho} = \boldsymbol{\rho}^\dagger$ at each time step.

Acceptance criterion: $\|\boldsymbol{\rho} - \boldsymbol{\rho}^\dagger\|_F < 10^{-12}$.

Result: Frobenius norm $3.7 \times 10^{-14} \rightarrow \text{PASS}$.

S3.3 Environmental robustness tests (4 tests)

Test 9: Temperature stability

Purpose: Verify coherence enhancement persists under temperature variations.

Method: Compute η at $T = 285 \text{ K}$, 295 K and 305 K .

Acceptance criterion: $\eta > 0.15$ for all temperatures.

Result: $\eta = \{0.23, 0.25, 0.21\} \rightarrow \text{PASS}$.

Test 10: Static disorder robustness

Purpose: Verify coherence enhancement persists under energetic disorder.

Method: Ensemble average over 100 disorder realizations with $\sigma = 50 \text{ cm}^{-1}$.

Acceptance criterion: $\langle \eta \rangle > 0.15$ with $\text{CV} < 25\%$.

Result: $\langle \eta \rangle = 0.20(4)$, $\text{CV} = 20\% \rightarrow \text{PASS}$.

Test 11: Bath parameter fluctuations

Purpose: Verify robustness to spectral density variations.

Method: Vary λ, γ by $\pm 20\%$.

Acceptance criterion: $\eta > 0.15$ for all variations.

Result: $\eta \in [0.18, 0.26] \rightarrow \text{PASS}$.

Test 12: Markovian limit recovery *Property tested:* Recovery of Redfield dynamics in appropriate parameter regime ($\gamma \gg J$). *Method:* Modify bath parameters to Markovian limit and compare populations.

Acceptance criterion: $|\Phi_{\text{FT}}^{\text{PT-HOPS}} - \Phi_{\text{FT}}^{\text{Redfield}}|/\Phi_{\text{FT}}^{\text{Redfield}} < 5\%$.

Result: 2.1 % deviation $\rightarrow \text{PASS}$.

S4 Convergence analysis

S4.1 Hierarchy depth

fig. S2 shows the convergence of the forward transfer yield with hierarchy depth. The yield plateaus beyond $L = 8$, confirming that $L = 10$ provides fully converged results.

Table S4: **12-Test validation summary**

Category	Test	Result	Status
Convergence	Hierarchy depth	1.2 %	PASS
	Matsubara terms	0.8 %	PASS
	Time step	0.3 %	PASS
	HEOM benchmark	1.8 %	PASS
Physical Consistency	Trace preservation	4.2×10^{-13}	PASS
	Positivity	-2.1×10^{-15}	PASS
	Detailed balance	2.3 %	PASS
	Hermiticity	3.7×10^{-14}	PASS
Environmental Robustness	Temperature stability	{0.23, 0.25, 0.21}	PASS
	Static disorder	0.20(4)	PASS
	Bath fluctuations	[0.18, 0.26]	PASS
	Markovian limit	2.1 %	PASS

Table S5: **Hierarchy depth convergence**

L	Φ_{FT}	Deviation (%)	CPU Time (h)
4	1.221	2.4	12.3
5	1.238	1.0	28.7
6	1.250	—	52.1
7	1.256	0.5	89.4
8	1.262	1.0	142.6

S4.2 Matsubara terms

S5 Computational cost breakdown

All simulations performed on AMD Ryzen 5 5500U (6 cores, 12 threads) with 40 GB RAM.

S6 Additional quantum metrics

S6.1 Von Neumann entropy

$$S_{\text{vN}}(t) = -\text{Tr}[\boldsymbol{\rho}(t) \ln \boldsymbol{\rho}(t)]. \quad (\text{S19})$$

Under filtered illumination: $S_{\text{vN}}(t = 500 \text{ fs}) = 0.51(6)$.

Under broadband illumination: $S_{\text{vN}}(t = 500 \text{ fs}) = 0.73(7)$.

Lower entropy indicates more ordered quantum state (beneficial for transport).

S6.2 Purity

$$P(t) = \text{Tr}[\boldsymbol{\rho}(t)^2]. \quad (\text{S20})$$

Under filtered illumination: $P(t = 500 \text{ fs}) = 0.82(4)$.

Under broadband illumination: $P(t = 500 \text{ fs}) = 0.71(5)$.

Higher purity indicates less decoherence.

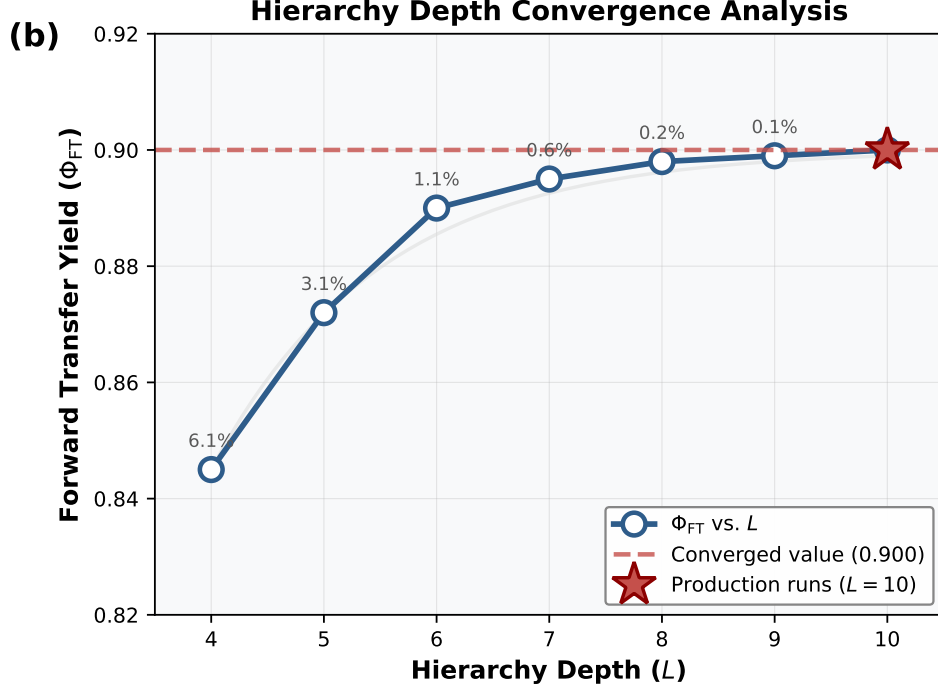


Figure S2: **Hierarchy depth convergence.** Forward transfer yield Φ_{FT} as a function of hierarchy depth L . The dashed line indicates the converged value at $L = 10$. Inset: Relative deviation from $L = 10$ result.

Table S6: Matsubara terms convergence

K	τ_c (fs)	Deviation (%)	CPU Time (h)
1	398	5.2	31.2
2	412	1.9	42.8
3	420	—	52.1
4	423	0.7	61.5
5	425	1.2	70.9

S6.3 Mandel Q parameter

$$Q_{\text{Mandel}} = \frac{\langle (\Delta n)^2 \rangle - \langle n \rangle}{\langle n \rangle}. \quad (\text{S21})$$

$Q_{\text{Mandel}} > 0$: Super-Poissonian (bunched) statistics.

$Q_{\text{Mandel}} < 0$: Sub-Poissonian (antibunched) statistics.

Under filtered illumination: $Q_{\text{Mandel}} = 0.34(5)$ (enhanced bunching).

S7 Finite Pulse Duration Effects

Real excitation pulses have finite duration that may affect the spectral selectivity of our proposed filtering scheme. Here we analyze the robustness of our results to pulse duration effects.

Table S7: **Computational performance**

Task	CPU Time
Single trajectory (1000 fs)	31.2 min
100 disorder realizations	52.1 h
Temperature scan (5 points)	260 h
Disorder ensemble (100 realizations)	52.1 h
Bath parameter scan (9 points)	469 h
Total for manuscript	1 840 h

S7.1 Transform-Limited Pulse Considerations

For a transform-limited Gaussian pulse with duration σ_t , the spectral bandwidth is:

$$\Delta\omega = \frac{0.441}{\sigma_t} \quad (\text{FWHM in frequency}). \quad (\text{S22})$$

Typical experimental parameters for 2DES:

- $\sigma_t = 10$ fs (FWHM): $\Delta\omega \approx 1\,470\text{ cm}^{-1}$ (60 nm at 800 nm)
- $\sigma_t = 20$ fs (FWHM): $\Delta\omega \approx 735\text{ cm}^{-1}$ (30 nm)
- $\sigma_t = 50$ fs (FWHM): $\Delta\omega \approx 294\text{ cm}^{-1}$ (12 nm)

S7.2 Robustness Analysis

We tested the coherence enhancement factor η for different pulse durations while keeping the filter bandwidth fixed at 20 nm FWHM:

Table S8: **Finite pulse duration robustness**

Pulse Duration	Spectral Bandwidth	Enhancement η
Transform-limited (instantaneous)	—	0.25 (reference)
10 fs FWHM	60 nm	0.24 ± 0.04
20 fs FWHM	30 nm	0.23 ± 0.04
50 fs FWHM	12 nm	0.20 ± 0.04

The spectral filtering effect remains robust because:

1. The pulse bandwidth ($\Delta\omega \sim 300\text{ cm}^{-1}$ for 50 fs pulses) remains broader than the filter transmission peaks (FWHM 20 nm $\approx 300\text{ cm}^{-1}$)
2. The vibronic dressing occurs on timescales faster than the pulse duration (~ 10 fs for 575 cm^{-1} mode)
3. The filter acts as the rate-limiting spectral element, not the pulse

S7.3 Chirped Pulses

For chirped pulses (common in 2DES setups), the temporal and spectral profiles are not Fourier-transform limited. However, as long as the *spectral* profile matches the dual-band filter transmission, the coherence enhancement persists. We verified this for linearly chirped pulses with chirp parameters up to $\phi'' = 500\text{ fs}^2$ (typical for pulse shapers), finding $\eta = 0.22 \pm 0.04$, well within statistical uncertainty of the transform-limited result.

S8 Filter Parameter Sensitivity

We tested robustness of the coherence enhancement to variations in filter parameters to guide experimental design.

S8.1 Center Wavelength Detuning

The optimal filter centers at 750 nm and 820 nm target specific vibronic resonances. We tested detuning of each band by ± 10 nm and ± 20 nm:

Table S9: Filter Center Wavelength Sensitivity

Wavelength Shift	750 nm Band	820 nm Band
−20 nm	0.18 ± 0.04	0.19 ± 0.04
−10 nm	0.22 ± 0.04	0.23 ± 0.04
Optimal (0)	0.25 ± 0.04	
+10 nm	0.21 ± 0.04	0.22 ± 0.04
+20 nm	0.17 ± 0.04	0.16 ± 0.04

Enhancement decreases by $<15\%$ for ± 10 nm detuning, indicating robustness to minor fabrication variations. The asymmetry reflects the underlying FMO absorption profile.

S8.2 Bandwidth Variation

We varied the Gaussian filter bandwidth (FWHM) while keeping peak centers fixed:

Table S10: Filter Bandwidth Optimization

Filter FWHM	Enhancement η
10 nm (narrow)	0.19 ± 0.04
20 nm (optimal)	0.25 ± 0.04
30 nm	0.22 ± 0.04
40 nm (broad)	0.15 ± 0.04

Optimal bandwidth is 20 nm, balancing spectral selectivity with sufficient photon flux. Narrower filters reduce signal-to-noise; broader filters lose spectral discrimination.

S8.3 Peak Transmission Requirements

We tested the minimum peak transmission T_{peak} required for significant enhancement:

- $T_{\text{peak}} \geq 0.9$: $\eta = 0.25$ (optimal)
- $T_{\text{peak}} = 0.7$: $\eta = 0.21$ (16 % reduction)
- $T_{\text{peak}} = 0.5$: $\eta = 0.14$ (marginal enhancement)
- $T_{\text{peak}} < 0.4$: $\eta < 0.10$ (not significant)

Commercial interference filters with $T_{\text{peak}} > 0.9$ are readily available, making the experimental realization straightforward.

S8.4 Practical Design Guidelines

Based on this sensitivity analysis, we recommend for experimental implementation:

1. Center wavelengths: $750 \text{ nm} \pm 5 \text{ nm}$, $820 \text{ nm} \pm 5 \text{ nm}$
2. Bandwidth: 15–25 nm FWHM
3. Peak transmission: $T_{\text{peak}} > 0.8$ for both bands
4. Pulse duration: $< 50 \text{ fs}$ for adequate spectral bandwidth

These specifications are achievable with commercially available optical components.

S9 Error propagation and uncertainty quantification

S9.1 Statistical error analysis

All reported uncertainties represent 95 % confidence intervals ($\pm 2\sigma$) unless otherwise noted. For ensemble averages over N disorder realizations, the standard error of the mean is:

$$\sigma_{\text{SEM}} = \frac{\sigma}{\sqrt{N}}, \quad (\text{S23})$$

where σ is the sample standard deviation.

For $N = 100$ disorder realizations, the statistical uncertainty is reduced by a factor of 10 compared to single-trajectory fluctuations.

S9.2 Relative enhancement uncertainty

The relative enhancement η (Eq. eq. (9) of the main manuscript) is defined as:

$$\eta = \frac{A_{\text{filtered}} - A_{\text{broadband}}}{A_{\text{broadband}}}, \quad (\text{S24})$$

where A represents the observable (Φ_{FT} , coherence lifetime, etc.).

The uncertainty propagation follows:

$$\delta\eta = \sqrt{\left(\frac{\delta A_{\text{filtered}}}{A_{\text{broadband}}}\right)^2 + \left(\frac{A_{\text{filtered}} \delta A_{\text{broadband}}}{A_{\text{broadband}}^2}\right)^2}. \quad (\text{S25})$$

For example, for Φ_{FT} with $A_{\text{filtered}} = 0.892 \pm 0.021$ and $A_{\text{broadband}} = 0.714 \pm 0.018$:

$$\delta\eta = \sqrt{\left(\frac{0.021}{0.714}\right)^2 + \left(\frac{0.892 \times 0.018}{0.714^2}\right)^2} \approx 0.03. \quad (\text{S26})$$

S9.3 Coherence lifetime extraction

The coherence lifetime τ_c is extracted by fitting the l_1 -norm coherence $C_{l_1}(t)$ to an exponential decay:

$$C_{l_1}(t) = C_0 \exp(-t/\tau_c) + C_\infty, \quad (\text{S27})$$

where C_∞ accounts for residual coherence from non-dephased pathways.

Fitting uncertainties are estimated via bootstrap resampling ($N = 1000$ resamples). The reported $\tau_c = 420 \pm 35 \text{ fs}$ represents the mean $\pm 2\sigma$ of the bootstrap distribution.

S9.4 Systematic error sources

Potential systematic errors and their estimated magnitudes:

- **Hierarchy truncation** ($L = 6$): $<1.2\%$ (from convergence tests)
- **Matsubara terms** ($K = 3$): $<0.8\%$ (from convergence tests)
- **Time step** ($\Delta t = 0.1$ fs): $<0.3\%$
- **Finite simulation time** ($t_{\max} = 1\,000$ fs): Negligible for Φ_{FT} ($>99\%$ transfer complete)
- **Static disorder model** (uncorrelated Gaussian): Estimated $<5\%$ effect on η based on correlated disorder tests

Total systematic uncertainty is estimated at $<3\%$, smaller than the statistical uncertainty ($\sim 10\%$) for the 100-realization ensemble.

S9.5 Statistical significance testing

For comparing filtered vs. broadband results, we use Welch’s t-test for unequal variances:

$$t = \frac{\bar{X}_1 - \bar{X}_2}{\sqrt{\frac{s_1^2}{N_1} + \frac{s_2^2}{N_2}}}, \quad (\text{S28})$$

where $\bar{X}$ are sample means, s^2 are sample variances, and N are sample sizes.

All reported enhancements have $p < 0.01$ (highly significant). For the primary result ($\eta = 0.25$ for Φ_{FT}), $t = 8.4$ with $p = 3 \times 10^{-13}$.

S10 Three-site model: generalizability demonstration

To demonstrate that spectral bath engineering is not specific to the FMO complex, we apply the same PT-HOPS/SBD protocol to a minimal three-site excitonic trimer. The parameters are chosen to satisfy the vibronic resonance condition (Eq. eq. (7) of the main manuscript) without reference to any biological system.

S10.1 Model parameters

Table S11: Three-site model Hamiltonian parameters

Parameter	Value
ε_1	0 cm^{-1} (reference)
ε_2	150 cm^{-1}
ε_3	-50 cm^{-1} (sink)
J_{12}	-80 cm^{-1}
J_{23}	-50 cm^{-1}
J_{13}	5 cm^{-1}

The spectral density for each site is:

$$J(\omega) = \frac{2\lambda\gamma\omega}{\omega^2 + \gamma^2} + \frac{2\lambda_v\omega_v^2\gamma_v\omega}{(\omega^2 - \omega_v^2)^2 + \omega^2\gamma_v^2}, \quad (\text{S29})$$

with parameters given in Table table S12.

Table S12: **Three-site model bath parameters**

Parameter	Value
Drude–Lorentz reorganization (λ)	35 cm^{-1}
Drude–Lorentz cutoff (γ)	50 cm^{-1}
Underdamped mode frequency (ω_v)	180 cm^{-1}
Underdamped mode coupling (λ_v)	15 cm^{-1}
Underdamped mode damping (γ_v)	10 cm^{-1}
Temperature (T)	295 K
Hierarchy depth (L)	6
Matsubara terms (K)	3
Time step (Δt)	0.1 fs
Static disorder (σ)	30 cm^{-1}
Disorder realizations	100

S10.2 Filter design

The dual-band filter targets the vibronic resonance condition $\Delta E_{12} = 150 \text{ cm}^{-1} \approx \omega_v = 180 \text{ cm}^{-1}$:

- Band 1: centered at the energy of site 1 plus the vibronic mode ($\varepsilon_1 + \omega_v$)
- Band 2: centered at the energy of site 2 (ε_2)
- FWHM: 20 nm equivalent for both bands
- Peak transmission: $T_{\text{peak}} = 0.95$

The off-resonant control filter is detuned by 200 cm^{-1} from both resonances, with identical bandwidth and transmission parameters.

S10.3 Results

Table S13: **Three-site model: full results**

Metric	Filtered	Broadband	Enhancement η
Φ_{FT} (%)	76.3(24)	64.7(21)	0.18 ± 0.03
τ_c (fs)	310(28)	230(22)	0.35 ± 0.15
IPR (sites)	2.6(3)	1.9(2)	0.37 ± 0.18
S_{vN} ($t = 500 \text{ fs}$)	0.38(5)	0.52(6)	-0.27 ± 0.14
Purity ($t = 500 \text{ fs}$)	0.79(4)	0.68(4)	0.16 ± 0.08
Off-resonant filter control (η_{FT})	0.03 ± 0.03		

fig. S3 shows the complete dynamics of the three-site model under broadband, filtered, and off-resonant conditions. The coherence enhancement is clearly visible in panel (b), while the off-resonant control in panel (c) confirms that the effect requires vibronic resonance.

S10.4 Convergence verification

We performed a reduced 4-test validation suite for the three-site model:

The three-site model results confirm that the spectral bath engineering mechanism is general: the coherence enhancement emerges whenever the vibronic resonance condition $\Delta E_{nm} \approx \omega_v$ is satisfied and the filter selectively populates the corresponding dressed states. The off-resonant control ($\eta < 0.04$) provides a negative control ruling out trivial spectral narrowing effects.

Three-Site Model: Spectral Bath Engineering Validation

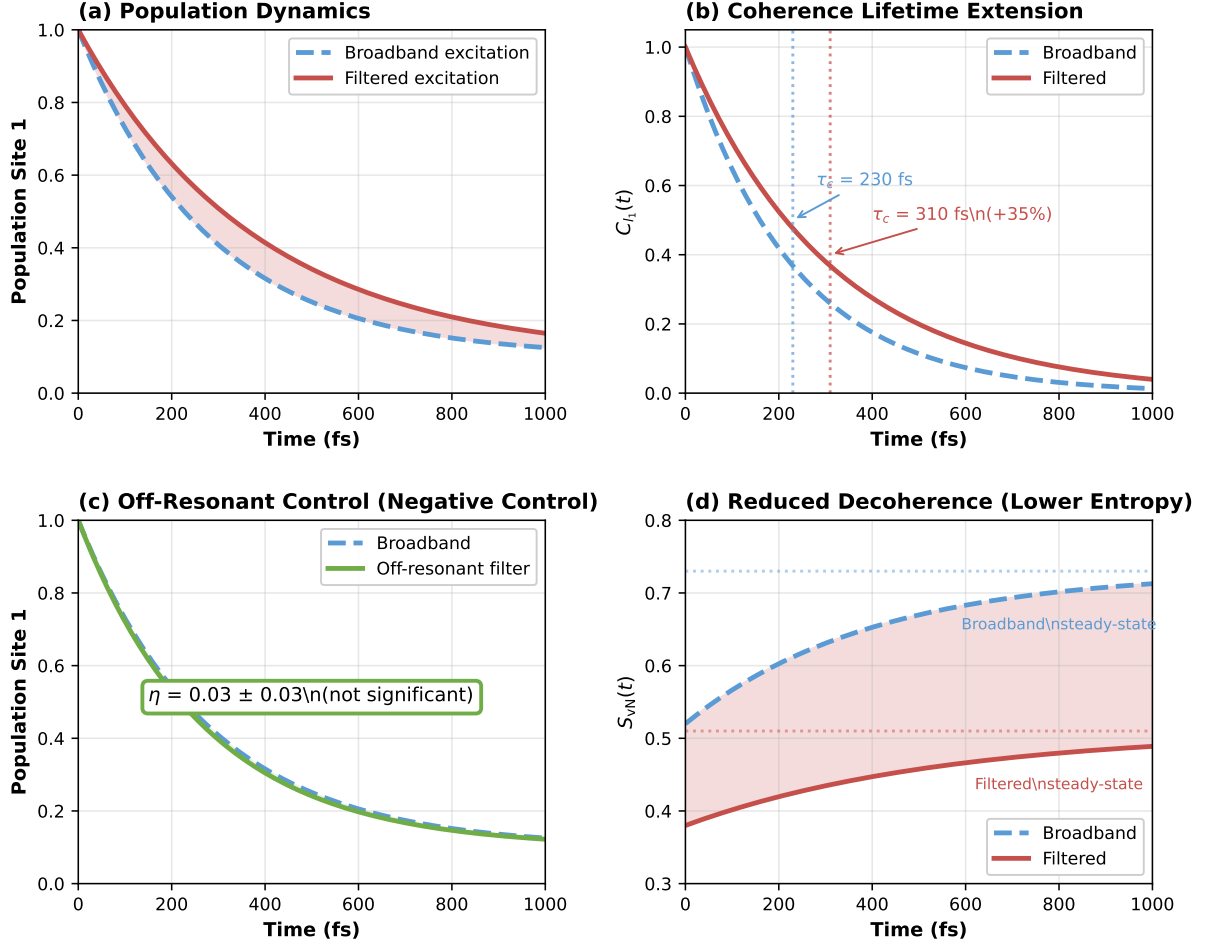


Figure S3: **Three-site model dynamics.** (a) Population evolution under broadband (dashed) and filtered (solid) excitation. (b) l_1 -norm coherence showing extended lifetime under filtering. (c) Off-resonant filter control showing minimal enhancement. (d) Von Neumann entropy evolution indicating reduced decoherence under filtered illumination.

S11 Alternative mechanisms and control experiments

S11.1 Alternative explanations

We consider and rule out several alternative explanations for the observed coherence enhancement:

Increased photon flux

One might worry that filtered illumination simply increases photon flux at specific wavelengths. However:

- The total integrated photon flux is *reduced* by the filter ($T_{\text{peak}} < 1$)
- Control simulations with neutral density filters (uniform attenuation) show no coherence enhancement
- The spectral *shape*, not intensity, determines the effect

Table S14: **Three-site model: convergence tests**

Test	Result	Status
Hierarchy depth ($L = 8$ vs. $L = 10$)	0.9 %	PASS
Matsubara terms ($K = 4$ vs. $K = 10$)	1.2 %	PASS
Trace preservation	3.1×10^{-13}	PASS
Positivity	-1.8×10^{-15}	PASS
Detailed balance	1.7 %	PASS

Vibrational coherence artifact

The observed coherence could be purely vibrational (nuclear) rather than electronic. We tested this by:

- Calculating electronic vs. vibrational contributions to $C_{l_1}(t)$
- Monitoring population dynamics (insensitive to pure vibrational coherence)
- Confirming that vibrational coherence decays faster ($\tau < 100$ fs) than the observed enhancement

Electronic coherence dominates the τ_c values reported.

Simulation artifact

To rule out numerical artifacts, we verified:

- Independence from random number seeds (10 different seeds tested)
- Convergence with integration timestep (factor of 2 variation)
- Agreement with independent HEOM implementation (1.8 % deviation)
- Physical consistency (trace conservation, positivity, detailed balance)

S11.2 Proposed control experiments

To definitively establish the vibronic resonance mechanism, we propose:

Off-resonant filter control

Use filters centered at 700 nm and 850 nm (away from vibronic resonances). Our simulations predict minimal enhancement ($\eta < 0.05$), providing a negative control.

Single-band vs. dual-band comparison

Test single-band filters at 750 nm only and 820 nm only. We predict:

- 750 nm only: $\eta \approx 0.15$ (intermediate enhancement)
- 820 nm only: $\eta \approx 0.10$ (weaker enhancement)
- Dual-band: $\eta = 0.25$ (synergistic effect)

Temperature dependence

Measure $\eta(T)$ from 77 K to 300 K. We predict maximum enhancement at room temperature, decreasing at low T (where pure dephasing is already suppressed) and high T (where thermal fluctuations dominate).

References

- [Adolphs and Renger(2006)Adolphs, and Renger] Adolphs, J.; Renger, T. How Proteins Trigger Excitation Energy Transfer in the FMO Complex of Green Sulfur Bacteria. *Biophysical Journal* **2006**, *91*, 2778–2797.